MEMO

To: Sarah Okafor, Head of E-Commerce **From:** Tung Nguyen, Data & ML team **Date:** 12 October 2026
Subject: The "also bought" box and our customer groups: results before Q4 planning

In short: we shrank the purchase history of 5,456 customers into a model **31 times smaller** than the table the old prototype builds. The whole nightly job now takes **under a second**. On purchases the model never saw, its suggestions caught what customers bought next for **40%** of them, against **27%** for showing best sellers. We also found four customer groups that marketing can plan around.

Which method we chose, and why

We use **truncated SVD** (singular value decomposition), a standard way to compress a large table. It describes every product and every customer with 70 numbers that capture buying patterns, such as "home decoration" or "children's items". Products with similar patterns are suggested together. It works directly on our purchase table, which is 98% empty. We also tested the main alternative, PCA. It gave the same suggestions (96% of lists identical) but needs an extra step and far more memory. All settings were chosen on an earlier period, so the results below are a fair test.

What we lose by compressing

The compressed model keeps about **a third of the detail** in the full table (35%). What it drops is mostly one-off choices that no other shop shares. The suggestions come from the shared patterns. That is why they generalise to products a shop has not bought yet, and why they still beat best sellers.

How big and how fast

Stored model: **5.1 MB** instead of 161 MB. The nightly job refits the model, writes a 10-product "also bought" list for each of 3,680 products and 10 suggestions for each customer. It takes **0.7 seconds** on a laptop and peaks at about 327 MB of memory.

Are the suggestions better than best sellers?

We hid the last two months (9 October to 9 December 2011) from the model. Then we checked whether any of a customer's 10 suggestions was a product they went on to buy for the first time. That happened for **40.4%** of 1,903 customers, against **26.9%** for the 10 best sellers they had not bought: **+13.5 points** (95% range +10.9 to +16.1). The suggestions also spread across 1,083 different products, while best sellers fill only 5.9% of the suggestion slots.

Customer groups for marketing

Group	Share	What they buy	Possible campaign
Big regular wholesalers	33%	Median 122 products over 7 orders; more than average: 72 sweetheart fairy cake cases, jumbo shopper vintage red paisley	Volume pricing and early access to new ranges
Home-decor shops	30%	Median 30 products over 2 orders; more than average: wood black board ant white finish, white hanging heart t-light holder	Seasonal decoration collections
Children's gift shops	26%	Median 27 products over 2 orders; more than average: red toadstool led night light, mini paint set vintage. 18% overseas (France, Germany, Belgium)	Back-to-school and kids' ranges; EU delivery offers
One-off small buyers	10%	Median 5 products over 1 order; few products in common	A second-order incentive after the first purchase

The groups come from the same model; the full picture is in the slides. Campaign ideas are suggestions for marketing to decide on.

What could go wrong

- **New customers and new products.** The model knows nothing about them, and 15% of purchases in the test period came from customers with no history. They should see best sellers until their first order.
- **Seasonality.** We tested on the pre-Christmas peak. Results in quieter months may differ, so the hit rate should be re-checked each month.
- **Popular products.** The model leans on what sells well, so some suggestions are items a shop would find anyway.
- **Data gaps.** 23% of order lines have no customer number and are not used.
- **Privacy.** Customer numbers never leave our team. This memo, the slides and the customer map use group names only.

Next steps

1. Run the box on product pages as a 4-week test against best sellers, measuring clicks and added revenue. 2. Schedule the nightly job and a monthly check of the hit rate. 3. Add a fallback for new products, such as items from the same range.